

INFO-H505 Projet report

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git link : <https://github.com/Kubixcube/smartstudy-cloud-agent>

1 Implementation Summary

Our implementation first enables automated PDF ingestion into MongoDB Atlas. To do this, the user uploads a PDF file to a Google Cloud Storage bucket. Once the upload is completed, the bucket emits an Object Finalize event, which triggers our Cloud Function `ingest_gcs_pdf`. This function downloads the PDF, extracts its text page by page, and then splits the extracted content into chunks. Each chunk is transformed into a vector embedding using Vertex AI, allowing us to represent the semantic meaning of each passage. Finally, the chunks, their metadata, and their corresponding vector embeddings are inserted or updated in MongoDB Atlas.

Our agent supports two main modes: a question-answering mode and a quiz mode.

In the question-answering mode, when the user asks a question, the system first determines whether the question targets a specific concept or asks for a broader document summary. If the question is about a specific topic, the question is converted into a vector embedding, and MongoDB Atlas Vector Search retrieves the chunks that are semantically closest to the query. If the user asks for a summary or an overview of the document, the system instead uses representative retrieval: it selects chunks distributed across the document in order to provide broader coverage.

In parallel, the system retrieves the recent conversation history from MongoDB. The final prompt is then built using the retrieved document context, the user question, the conversation history, and the agent-specific instructions. These instructions require the agent to answer pedagogically, cite its sources, and stay grounded in the retrieved course material. The prompt is sent to Gemini 2.5 Flash, which generates the final answer. The answer is displayed to the user and then stored in MongoDB, together with the user question, so that it can be reused as conversation memory in later turns.

In quiz mode, the user provides a topic or quiz request. The topic is converted into a vector embedding using Vertex AI, and MongoDB Atlas Vector Search retrieves the most relevant chunks from the indexed lecture material. A quiz-specific prompt is then built using this retrieved context and instructions requiring Gemini to generate a five-question quiz, include correct answers, and cite the source file and page used for each answer. Gemini 2.5 Flash is then used to generate the final quiz.

The most challenging part of moving from the labs to the real cloud environment was having to set up the database, the storage bucket, and the other cloud resources ourselves. This took a significant amount of time because we encountered many errors.